

# The Tutor

An on-device teaching agent and the tool layer (MCP) that lets it draw simulations, write notes, build slides and docs, and teach anything with flash cards, recall and memory tricks — in under 2 GB of RAM.

|           |                                                                                                                                                                                                                 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Written   | 10 October 2026                                                                                                                                                                                                 |
| For       | Rohan Singh (product owner) and any developer joining SIMBLIP                                                                                                                                                   |
| Built by  | Claude, developing inside this repository, with Rohan steering                                                                                                                                                  |
| Companion | Document 2 — “The Sketch Model” (drawing & visuals → simulation)                                                                                                                                                |
| Basis     | Read directly from the repo at commit <code>979bc08</code> : <code>lib/ai/*</code> , <code>app/api/ai</code> , <code>lib/scene/*</code> , <code>lib/store/*</code> ,<br>course mode, walkthrough engine, docs/* |

# How to read this document

This is one of two documents. It covers the **Tutor**: a language model plus a set of tools. Document 2 covers the **Sketch model**: a vision model that turns drawings and pictures into simulations. They share one runtime (Section 5), so read Section 5 here even if you only work on sketching.

## IF YOU HAVE 10 MINUTES

Read Section 1 (what we make), the “decisions” table in Section 3, and the phase list in Section 7.

## IF YOU ARE ABOUT TO WRITE CODE

Read Section 2 (what already exists — reuse it), then Section 4 (tool layer), then the phase you are on in Section 7.

## IF YOU ARE ROHAN

Section 8 is your learning path. Section 9 lists what is still unmeasured and the decisions that are yours.

## IF YOU ARE CLAUDE

Section 7 is the work order. Every phase has files, tests and an exit criterion. Do not start a phase before the previous exit criterion is met.

## Contents

- |   |                                          |   |                                                 |
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| 1 | What we are making                       | 6 | The teaching engine: recall, flash cards, hints |
| 2 | What already exists in SIMBLIP           | 7 | How to build it: phases, files, exit criteria   |
| 3 | Architecture and the decisions behind it | 8 | What you need to learn, and the tool list       |
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| 5 | The brain: model, memory budget, runtime | A | Appendix: tool catalogue, glossary, sources     |

### HONESTY ABOUT NUMBERS

Figures marked **est.** are planning estimates from model architecture arithmetic or vendor claims, not measurements on a SIMBLIP build. Phase 0 exists to replace each of them with a measurement. Figures marked **measured** come from comments in this repo (RTX 4060 8 GB, the author’s machine). Facts about outside projects (WebGPU, WebMCP, SmoIVLM, WebLLM) came from a web search on 10 Oct 2026 and are listed in the Appendix; where sources disagreed, this document says so.

# 1. What we are making

## 1.1 One sentence

A teacher that lives inside the notebook, runs on the student’s own device, and teaches by *doing* — it builds a simulation you can poke, asks you to predict what will happen, then turns what you just learned into flash cards that come back at the right time.

## 1.2 The three things you asked for, restated as products

| # | Product                   | In plain words                                                                                                                                        | Document                           |
|---|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1 | <b>The Tutor + Tools</b>  | A model that can call every action SIMBLIP has — draw a circuit, run it, write notes, make slides, quiz you — through a safe, typed tool layer (MCP). | This one                           |
| 2 | <b>The Sketch model</b>   | You draw a ramp and a ball, or photograph a circuit from a textbook, and it becomes a live simulation.                                                | Document 2                         |
| 3 | <b>On-device shipping</b> | Both run on the client: $\leq 2$ GB RAM, a few CPU cores, no server GPU, nothing about a student leaves their device.                                 | Section 5 here; Section 6 of Doc 2 |

## 1.3 Who it is for — and what “natural” has to mean

SIMBLIP’s audience is university engineering students and their teachers, many of them in places where a laptop has 4–8 GB of RAM and no discrete GPU, and where “technical” and “non-technical” students sit in the same room. So “natural” is a design constraint, not a slogan. It means:

- **No prompt-engineering required.** A student can type “I don’t get why a pendulum slows down” or drop a photo of an exam question. A teacher can type “slides on diode rectifiers for tomorrow, 8 slides, with a quiz”.
- **Show before tell.** Wherever a concept has a simulation (and SIMBLIP has engines for mechanics, circuits, optics, waves, quantum, DSA and 70 step-engines), the tutor builds it first and explains from it.
- **Nothing happens behind your back.** The assistant never edits silently. Every change is previewed, applied as *one* undo step, and shows what it touched (this rule already exists in the repo and must survive).
- **It adapts to the person.** Same concept, three voices: “new to this”, “I know the basics”, “exam in 3 days”. Language level, analogies and how much maths are different for each.
- **It is honest.** A 1–2 B-parameter model is wrong sometimes. The product answer is grounding (course content and engines are the source of truth) and verification (lint, engine run, arithmetic check) — not confidence.

## 1.4 What a session looks like

| Student says                     | The Tutor does (tools in monospace )                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Teach me projectile motion.”    | <code>learn.diagnose</code> → one question to find the level. <code>course.search</code> finds the lesson.<br><code>scene.build</code> places a cannon + ball + graph, previewed. <b>Predict:</b> “If I double the angle, does the range double?” — student picks. <code>scene.run</code> , then <code>scene.measure</code> reads the real result. Tutor explains the gap. <code>learn.cards.create</code> makes 3 cards (one links back to this very simulation).<br><code>learn.schedule</code> sets review. |
| (photo of a circuit from a book) | <code>sketch.recognize</code> (Doc 2) → scene graph → <code>scene.build</code> . “Here is your circuit. Want to see the current through R2?” <code>scene.run</code> → <code>graph.plot</code> .                                                                                                                                                                                                                                                                                                                |
| “I have a test tomorrow.”        | <code>learn.mastery.get</code> → weakest 5 concepts. <code>learn.quiz.generate</code> (retrieval practice, mixed order). Wrong answers → <code>learn.hint</code> ladder, never the bare answer first. <code>learn.mnemonic</code> for facts that need rote.                                                                                                                                                                                                                                                    |

|                                                                                      |                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teacher: "Make a worksheet and slides for Kirchhoff's laws, easy and hard versions." | <code>course.search</code> → <code>slides.from_outline</code> + <code>doc.worksheet</code> (two difficulty levels) → preview both → <code>teacher.assignment.create</code> (needs teacher role) → class dashboard later shows which questions failed. |
| "Explain this line" (selected text in a lesson)                                      | Already works today as <code>askAboutSelection</code> — the passage arrives as an attachment. It stays, and gains the hint ladder.                                                                                                                    |

## 1.5 Non-goals (so scope stays honest)

- Not a general chatbot. It will decline to be your essay-writer; it exists to teach what SIMBLIP can show.
- Not a replacement for authored courses. The memory file for Course Mode records the decision: *lessons are authored offline and lint-gated; runtime stays dumb*. The tutor **retrieves and adapts** authored content — it does not invent a lesson and present it as the syllabus.
- Not cloud-dependent. A hosted fallback is allowed (Section 3), but the product must be fully useful offline.
- Not a 7B-model product. The existing 7B setup proved the pipeline; the shipped brain is 0.6–1.7 B parameters and has to be *made* good by training and verification, not by size.

## 1.6 Definition of success

| Dimension             | Target                                                                                                                                           | How it is measured                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Footprint             | AI process ≤ 2 GB peak, Lite tier ≤ 1 GB                                                                                                         | Phase 2 harness on 3 reference devices (Section 5.5)                    |
| Correctness of scenes | ≥ 80% lint-clean on the held-out course prompts with <b>no few-shot examples</b> (today a 7B + 4 examples reaches 12/15 = 80%, <b>measured</b> ) | <code>scripts/ai-sweep.mjs</code> against the engine                    |
| Tool-call validity    | ≥ 98% schema-valid (guaranteed by grammar-constrained decoding), ≥ 90% <i>correct tool</i>                                                       | Tool eval set (Section 9)                                               |
| Latency (Mid device)  | First useful token ≤ 3 s warm; a scene ≤ 15 s                                                                                                    | Phase 2 harness                                                         |
| Learning              | Students retain more at 7 days with the tutor loop than with notes alone                                                                         | A real classroom pilot (Section 9) — the only metric that truly matters |

## 2. What already exists in SIMBLIP

Most of this project is **connecting things that already exist**. Before writing any code, know this map. Everything below was read in the repo, not assumed.

### 2.1 The core idea you must not break

`docs/architecture.md` states the founding rule: *the simulation engine is the product; the notebook is the environment*. Every object is **geometry + behaviors** (`SceneObject` in `lib/scene/types.ts`); “the AI is only an assistant — it returns the same primitives and behaviors a user could place by hand, validated against a schema, imported through ordinary store actions. The engine never depends on it.” Both new models inherit this contract. **Neither model gets a private way to change the canvas.**

### 2.2 The AI pipeline today

| Piece         | File                                                                                                       | What it does                                                                                                                                                                                                                              | What we do with it                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Gateway       | <code>app/api/ai/route.ts</code>                                                                           | One request → intent routing → one lane → verified result. Optional SSE streaming. Redis cache wrapper.                                                                                                                                   | Keep as the <i>hosted</i> backend. Logic moves behind an engine interface so the browser can run it.                                |
| Intent router | <code>lib/ai/route-intent.ts</code>                                                                        | Pure regex: <code>simulate · explain · both · edit</code> . No model call. Edit only fires when a page/selection is attached.                                                                                                             | Extend with new lanes (teach, recall, quiz, slides, sketch). Cheap and testable — keep it model-free.                               |
| Simulate lane | <code>generate.ts</code> , <code>simscrip-corporus.ts</code> , <code>few-shot.ts</code>                    | SimScript generation. ~1,021-token system prompt, retrieved few-shots, lint, auto-fix, up to 2 repairs. A 103-tool agent was <i>removed</i> because its tools alone cost 17,981 prompt tokens ( <b>measured</b> : 11.9 s/round vs 2.2 s). | Becomes the SimScript tool. The prompt-size lesson drives the whole tool-layer design (Section 4).                                  |
| Verifier      | <code>simscrip-lint.ts</code> , <code>simscrip-fix.ts</code> , <code>verify-scene.ts</code>                | Static checks (unknown kind, missing anchor, bad channel, nonsensical behavior) and deterministic repairs.                                                                                                                                | <b>The most valuable asset.</b> “Verification is free, generation is not.” It becomes the reward signal for training (Section 5.6). |
| Explain lane  | <code>explain.ts</code> , <code>arithmetic.ts</code>                                                       | Markdown + LaTeX answers split into notebook objects. An arithmetic checker re-computes numbers the model wrote. System prompt already says “Nepali university engineering notebook”.                                                     | Base of the explain/teach tools. Arithmetic checker generalises to a “claims checker”.                                              |
| Edit lane     | <code>edit-ops.ts</code> , <code>edit.ts</code> , <code>apply-ops.ts</code> , <code>page-context.ts</code> | JSON op list ( <code>add · update · remove · setParam · addBehavior · setVariable</code> ), verified against the real page, applied as <b>one</b> history entry. Page digest capped at 4,000 chars.                                       | Becomes <code>scene.edit</code> . The digest is the model's “eyes” on the page.                                                     |
| Slides        | <code>slides.ts</code>                                                                                     | Markdown answer → pptx-kind page, one slide per heading, overflow split across “continued” slides.                                                                                                                                        | Becomes <code>slides.from_outline</code> .                                                                                          |

|             |                                                                 |                                                                                 |                                                                      |
|-------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Attachments | <code>attachments.ts</code>                                     | PDF/pptx/xlsx → text in the browser; images → Tesseract OCR (lazy-loaded WASM). | Already client-side — proof that heavy WASM tooling fits the app.    |
| Panel       | <code>components/workspace/ai-panel.tsx</code>                  | 1,212 lines: model picker, Auto/Manual, context chips, attachments, streaming.  | Gains a “mode” (Learn / Answer) and a Recall tab. Do not rebuild it. |
| Chat state  | <code>lib/store/ai-chat.ts</code> , <code>ai-sessions.ts</code> | Per-user persisted thread, 30-turn cap, last 3 turns sent.                      | Keep. Learner state goes in a <i>separate</i> store (Section 6.5).   |

## 2.3 Other parts the Tutor will stand on

| Part                               | Where                                                                                                                                                                                        | Why it matters to the tutor                                                                                                                                                   |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Store actions with history         | <code>lib/store/document.ts</code> ( <code>addObject</code> , <code>updateObject</code> , <code>setParam</code> , <code>pushHistory</code> ... )                                             | The only mutation path. Tools wrap these; nothing new needed in the store.                                                                                                    |
| Component registry                 | <code>lib/scene/factory.ts</code> ( <code>COMPONENTS</code> )                                                                                                                                | The model names a component id; it never hand-builds a <code>SceneObject</code> .                                                                                             |
| SimScript + <code>diagram()</code> | <code>lib/scene/simscript.ts</code> , <code>diagram.ts</code> , <code>simscript-component-reference.md</code>                                                                                | Dense, high-level scene language; <code>diagram()</code> will draw concept maps for free.                                                                                     |
| Engines                            | <code>lib/physics</code> , <code>circuit</code> , <code>optics</code> , <code>waves</code> , <code>quantum</code> , <code>dsa</code> , <code>econ</code> , <code>steplab</code> (70 engines) | The ground truth. The tutor can <i>measure</i> a running simulation ( <code>scene.measure</code> ) to check a student's prediction.                                           |
| Course Mode                        | <code>lib/store/course.ts</code> , <code>content/courses/**</code> , <code>/api/courses</code>                                                                                               | 20 authored computer-engineering courses plus the syllabus plan across 5 subjects. Lessons already contain MCQs, problems and steps — the retrieval corpus and the quiz seed. |
| Walkthrough engine                 | <code>lib/walkthrough/*</code>                                                                                                                                                               | A cursor-driven engine that moves a visible cursor and narrates. Reusable as “show me where” — a tutor that points.                                                           |
| Roles                              | <code>lib/auth/types.ts</code> ( <code>super_admin</code> · <code>admin</code> · <code>teacher</code> · <code>student</code> · <code>board</code> , <code>can(role, permission)</code> )     | Tool permissions reuse this — no second permission system.                                                                                                                    |
| Assignments and insights           | <code>components/workspace/assignments-panel.tsx</code> , <code>insights-dashboard.tsx</code>                                                                                                | The teacher-facing sink for quiz results.                                                                                                                                     |
| Storage + sync                     | <code>lib/storage/opfs.ts</code> , <code>quota.ts</code> , <code>lib/sync/*</code> , <code>/api/sync</code>                                                                                  | OPFS holds model weights; sync carries learner state across devices.                                                                                                          |
| Test harness                       | <code>*.test.mjs</code> next to code, <code>scripts/test/register.mjs</code> alias loader                                                                                                    | Plain Node tests, no browser. New pure modules must follow this shape.                                                                                                        |

## 2.4 What does *not* exist yet (the real gaps)

1. **No on-device inference.** Every model call goes to `OLLAMA_HOST` (default `localhost:11434`) or OpenRouter. On Vercel nothing can reach a local Ollama — the deployment-plan memory already lists “local-Ollama-in-prod” as a pre-launch fix. **On-device inference solves a launch blocker, not just a feature.**
2. **No learner model.** Nothing records what a student knows, got wrong or should review. No flash cards, no spaced repetition (a search of the repo found “recall” only in UX planning docs).
3. **No tool registry.** The old `lib/ai/tools.ts` was deleted with the 103-tool agent; today “what the AI can do” is spread across lanes. There is no single typed list of actions.
4. **No MCP surface.** Nothing lets an outside agent (including Claude Code) drive the app.

5. **Text-only.** Every local model is text-only; an image becomes OCR text ( `attachments.ts` ), so a circuit photo yields almost nothing. This is the gap Document 2 fills.
6. **The tutor persona is a one-line system prompt** ("engineering tutor ... Nepali university engineering notebook"). No levels, no hint policy, no languages.

# 3. Architecture and the decisions behind it

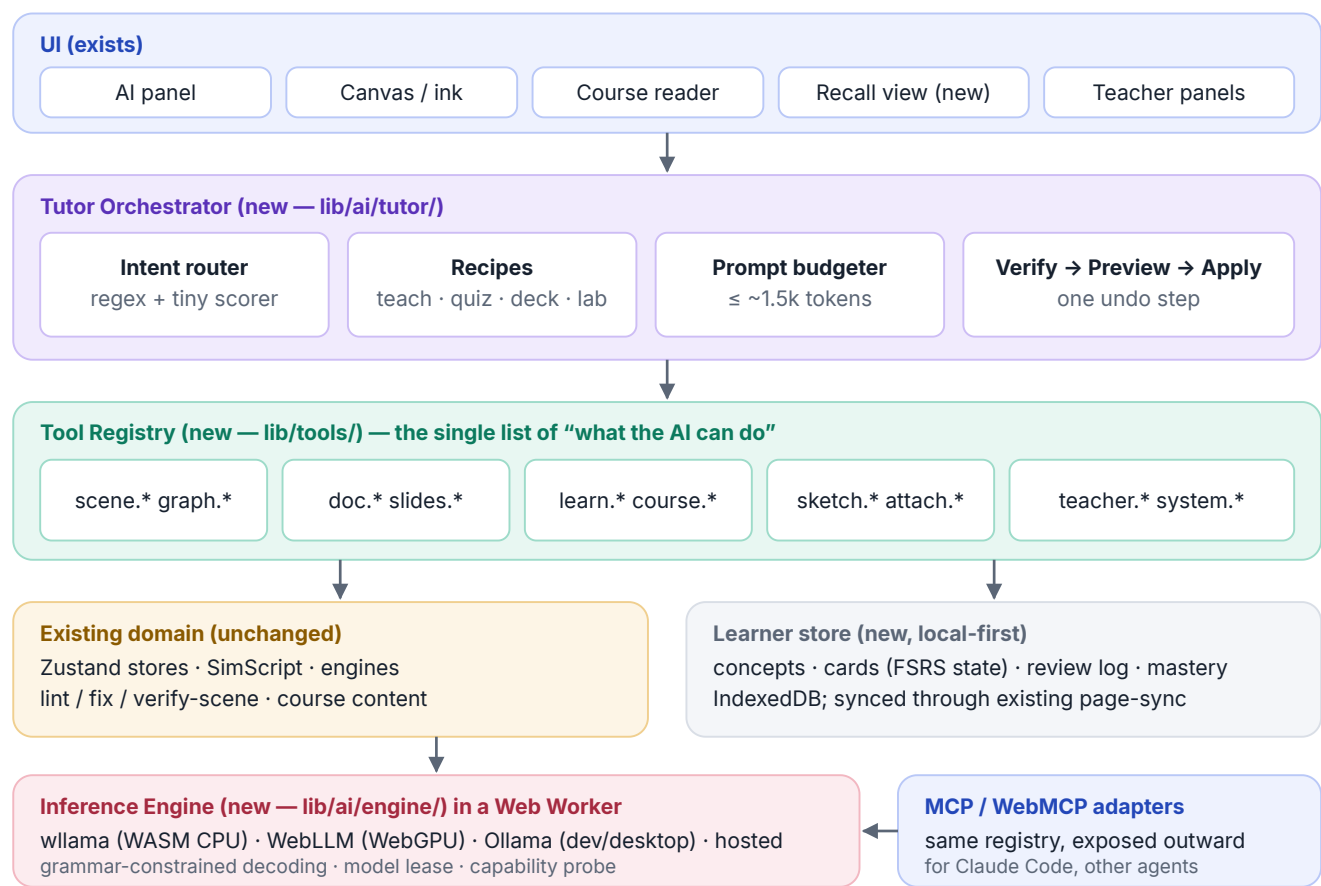


Figure 1 — Layers. Only the boxes marked “new” are built; everything else is wrapped, not rewritten.

## 3.1 The decisions, and why

| #  | Decision                                                                                                                                                                                                                                                | Why (and what we rejected)                                                                                                                                                                                                                   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1 | <b>Workflows (“recipes”) over free-roaming agents.</b> A recipe is a fixed sequence of tool calls where the model only fills slots (topic, difficulty, wording). Example: <i>teach</i> = diagnose → retrieve → build → predict → run → explain → cards. | A 1–2 B model loses the thread in open-ended ReAct loops; the repo already learned this with the 24-round agent. Recipes make behaviour testable and cap cost at 1–3 generations per turn. <i>Rejected</i> : general agent loop.             |
| D2 | <b>A tool registry is the single source of truth;</b> lanes, the local model, WebMCP and an MCP server are all <i>adapters</i> over it.                                                                                                                 | The old code had three hand-maintained copies of “what is placeable” and they drifted ( <code>schema.ts</code> supported 9 of ~22 kinds). One registry, derived from <code>COMPONENTS</code> and <code>BEHAVIOR_TYPES</code> , cannot drift. |
| D3 | <b>MCP is the outer contract, not the hot path.</b> The on-device model calls registry functions directly in-process. MCP/WebMCP expose the same functions to outside agents.                                                                           | JSON-RPC round trips and long tool descriptions cost tokens and time a small model cannot spare. MCP buys interoperability (and lets Claude Code drive and test the app), not speed.                                                         |
| D4 | <b>The model sees ≤ 6 tools per turn,</b> chosen by the intent router; never the whole catalogue.                                                                                                                                                       | Direct lesson from <span>measured</span> history: 103 tools = 17,981 tokens. A 40-tool catalogue shown whole would still be ~7k tokens ( <span>est.</span> ) — fatal on CPU where prefill is slow.                                           |
| D5 | <b>One brain, fine-tuned for all lanes</b> (tool calls, SimScript, explanations, edits, card writing), not one model per lane and not LoRA hot-swapping in v1.                                                                                          | RAM: two resident models would not fit in 2 GB. One multi-task fine-tune is the simplest thing that fits. Adapters can be revisited if one task regresses another.                                                                           |

|     |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                             |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D6  | <b>Fine-tune to remove few-shot prompts.</b>                                                                                                                                                                                   | On CPU, prefill (reading the prompt) is the slow part. Today's pipeline puts a ~1k-token system prompt <i>plus</i> 4 example scripts in front of every request. Baking that into the weights shrinks the prompt to ~150–300 tokens ( <b>est.</b> ) — the largest single speed-up available. |
| D7  | <b>Grammar-constrained decoding for every structured output.</b>                                                                                                                                                               | Tool-call JSON and edit ops become valid <i>by construction</i> (llama.cpp GBNF in wllama; JSON-schema grammar in WebLLM). Small models fail at syntax first; this removes that failure class and most repair rounds.                                                                       |
| D8  | <b>The verifier is the safety net and the teacher.</b><br>Nothing reaches the canvas unlited; the same lint + engine-run is the training reward.                                                                               | Keeps “the AI is only an assistant” true, and means training data can be machine-checked at scale.                                                                                                                                                                                          |
| D9  | <b>Learner state is local-first and separate from chat.</b>                                                                                                                                                                    | A student's mistakes are sensitive and the audience includes minors. Default: nothing leaves the device. A teacher sees aggregates only with an explicit class-join consent.                                                                                                                |
| D10 | <b>Pedagogy runs without the model where possible.</b><br>Spaced-repetition scheduling, mastery, hint-ladder state and quiz grading are deterministic code; the model only <i>writes</i> content (a card, a hint, an analogy). |                                                                                                                                                                                                                                                                                             |
| D11 | <b>Hosted fallback stays</b> for devices that cannot run the model (< 4 GB, no WASM threads).                                                                                                                                  | The current <code>/api/ai</code> keeps working; the engine selects a backend by capability probe. Never block a student for hardware.                                                                                                                                                       |

## 3.2 One request, end to end

```

"Teach me projectile motion"                                (student, Learn mode, board page open)
1  router.classify()      → lane = teach      (regex; 0 ms; no model)
2  budgeter.assemble()    → system(≈250 tok) + page digest(≤4k chars) + learner snapshot(≈60 tok)
3  recipe "teach" step A  → model emits {tool:"learn.diagnose", args:{topic:"projectile motion"}}
                           (grammar-constrained → always valid JSON)
4  registry.run(...)      → returns one question; UI shows it; student answers (no model time spent)
5  recipe step B          → course.search("projectile") → BM25 over lessons → lesson excerpt (≈400
tok)
6  recipe step C          → model emits SimScript for the scene → lint → autofix → ok
7  Preview               → ghost scene + "Add to canvas" (Auto mode applies immediately)
8  recipe step D          → "Predict: what happens to range if the angle doubles?" (UI, not model)
9  student picks; scene.run(); scene.measure() → real number → tutor compares → explains
10 recipe step E          → model writes 3 cards (grammar: {front, back, hint}) → learn.cards.create
11 learner store          → FSRS schedules first review; Recall tab badge updates

```

Only steps 3, 6, 9 (explanation) and 10 use the model, and each is short. That is deliberate: **the CPU spends its time on the few outputs only a model can produce.**

### 3.3 Where code goes (new folders, existing conventions)

```
lib/tools/          registry.ts · types.ts · scene/ doc/ slides/ learn/ course/ sketch/ teacher/
system/

lib/ai/engine/     adapters/ local.ts (grammar) · webmcp.ts · mcp-server.ts
                  types.ts (InferenceEngine) · probe.ts · worker.ts · wllama.ts · webllm.ts
                  ollama.ts · hosted.ts · lease.ts · download.ts · grammar.ts

lib/ai/tutor/      router.ts · recipes/ · budget.ts · persona.ts · hints.ts · verify.ts
lib/learning/      store.ts (zustand + IDB) · fsrs.ts · cards.ts · mastery.ts · concepts.ts ·
mnemonic.ts

components/workspace/ recall-view.tsx · learn-mode-toggle.tsx · model-manager.tsx
scripts/ai/        data-factory/ · train/ · eval/ · export/
docs/ai/           decisions.md (decision log; append, never rewrite)
```

Naming follows the repo: kebab-case files, `*.test.mjs` beside pure modules, `'use client'` only where React state is touched. Pure modules must stay importable from Node tests and from the browser — the lint/explain modules already honour this, and the Redis cache is deliberately wrapped outside them (see the comment in `app/api/ai/route.ts`).

# 4. The tool layer (MCP)

## 4.1 What MCP is, and how we use it

The Model Context Protocol standardises how an AI client discovers and calls an application’s capabilities. A server offers three things: **tools** (actions with typed inputs), **resources** (readable data), and **prompts** (reusable templates). For SIMBLIP:

| MCP primitive | SIMBLIP mapping                | Example                                                                           |
|---------------|--------------------------------|-----------------------------------------------------------------------------------|
| Tool          | A registry entry               | scene.build , learn.cards.create                                                  |
| Resource      | Read-only views                | simblip://page/{id}/digest , simblip://course/{id} ,<br>simblip://learner/mastery |
| Prompt        | The recipes, as slash-commands | /teach topic , /quiz me , /deck from page                                         |

Three delivery paths for the same registry:

- 1. **In-process (the on-device model)**. Direct function call, schema from the registry, grammar built from it. Zero protocol overhead. Primary
- 2. **WebMCP (in the page)**. A draft browser API that lets a page publish tools to an agent in the browser; the search found an early preview in Chrome 146 Canary behind a flag and a still-moving specification. Treat as optional and polyfilled ( @mcp-b/\* packages wrap the official SDK for browsers). Draft spec
- 3. **MCP server (outside agents)**. A Streamable-HTTP endpoint ( /api/mcp ) that relays tool calls to the user’s open tab over the realtime channel the app already has ( board-live , Redis pub/sub). This is what lets Claude Code or Claude Desktop drive a running notebook. Authenticated as the user, off by default, scoped by role. The official TypeScript SDK v2 was in beta in the search results (server and client split into separate packages); check whether stable shipped before pinning a version. Developer + power-user feature

### WHY BOTHER WITH MCP AT ALL

Because it turns SIMBLIP into something an agent can *operate and test*. Phase 2 below uses it so that Claude, while developing, can open a page, run a tool, read the digest and judge the result — the loop that produces the training data in Section 5.6.

## 4.2 What a tool is (the contract)

```
// lib/tools/types.ts
export type Effect = 'read' | 'write' | 'run' // write = changes a page; run = starts/stops engines
export interface Tool<I, O> {
  name: string // 'scene.build' - dotted, stable, never renamed
  summary: string // ≤ 12 words. This is all the small model reads.
  help?: string // long text for MCP clients and humans, never put in a small-
model prompt
  input: z.ZodType<I> // single source of truth → JSON Schema (MCP) AND grammar (local
model)
  effect: Effect
  needs?: Permission // from lib/auth/types.ts, e.g. 'assignment:create'
  confirm?: 'preview' | 'always' | 'never' // write tools default to 'preview'
  run(args: I, ctx: ToolContext): Promise<ToolResult<O>>
}
interface ToolContext { pageId: string; role: Role; selection: string[]; signal: AbortSignal; engine:
InferenceEngine }
interface ToolResult<O> { ok: boolean; data?: O; error?: string; repairHint?: string; undoToken?: string }
```

## Six rules that keep the layer small-model-friendly

1. **Names are verbs the model has seen.** `scene.build`, not `createSimulationFromNaturalLanguage`.
2. **Summaries  $\leq 12$  words; arguments  $\leq 5$ , flat, mostly enums and short strings.** Nested objects are where 1 B models fall over.
3. **Errors carry a `repairHint`** in the model's own vocabulary ("unknown component 'switch2'; did you mean 'switch'?"). This reuses the generate  $\rightarrow$  verify  $\rightarrow$  repair loop that already caps at 2 repairs ( `MAX_REPAIRS` ).
4. **Write tools never write immediately.** They return a *proposal* (the verified script or op list). Apply is a separate, user-gated step that calls `pushHistory` once and runs ops with `{history:false}` — the exact guarantee in `docs/ai-direct-edit-plan.md`.
5. **Re-verify at apply time.** The page can change between proposal and apply; stale ids abort the whole batch. Never apply half a patch.
6. **Roles are checked in the registry, not in prompts.** A student asking for `teacher.assignment.create` gets a refusal from code, whatever the model says.

## 4.3 The catalogue (v1, 42 tools)

Full table in Appendix A. Summary by family; "shown when" is what the intent router uses to keep each turn to  $\leq 6$  tools.

| Family                 | Tools                                                                                                                                                                                                                                                                                                                             | Effect         | Wraps (existing)                                                                                                                                        |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>scene.*</code>   | <code>describe</code> · <code>build</code> · <code>edit</code> · <code>add</code> · <code>connect</code> · <code>set_param</code> · <code>run</code> · <code>pause</code> · <code>reset</code> · <code>measure</code> · <code>snapshot</code>                                                                                     | read/write/run | <code>generate.ts</code> , <code>edit-ops.ts</code> , <code>apply-ops.ts</code> , <code>transport/physics runtime</code> , <code>page-context.ts</code> |
| <code>graph.*</code>   | <code>plot</code> · <code>bind</code> · <code>compare</code>                                                                                                                                                                                                                                                                      | write          | <code>SimScript graph.plot</code> , <code>graph objects</code>                                                                                          |
| <code>doc.*</code>     | <code>note</code> · <code>outline</code> · <code>formula</code> · <code>table</code> · <code>worksheet</code> · <code>summarise</code>                                                                                                                                                                                            | write          | <code>explain.ts</code> , <code>text engine (Tiptap)</code> , <code>table-data.ts</code>                                                                |
| <code>slides.*</code>  | <code>from_outline</code> · <code>add_slide</code> · <code>export</code>                                                                                                                                                                                                                                                          | write          | <code>slides.ts</code> , <code>pptx-export.ts</code>                                                                                                    |
| <code>course.*</code>  | <code>search</code> · <code>open_lesson</code> · <code>walkthrough</code>                                                                                                                                                                                                                                                         | read/run       | <code>/api/courses</code> , <code>lib/walkthrough</code>                                                                                                |
| <code>learn.*</code>   | <code>diagnose</code> · <code>hint</code> · <code>quiz.generate</code> · <code>quiz.grade</code> · <code>cards.create</code> · <code>cards.next</code> · <code>cards.grade</code> · <code>mastery.get</code> · <code>plan</code> · <code>mnemonic</code> · <code>concept_map</code> · <code>predict</code> · <code>feynman</code> | read/write     | <i>new</i> (Section 6), <code>diagram.ts</code> for maps                                                                                                |
| <code>sketch.*</code>  | <code>recognize</code> · <code>to_scene</code>                                                                                                                                                                                                                                                                                    | read/write     | Document 2                                                                                                                                              |
| <code>attach.*</code>  | <code>read_file</code> · <code>ocr</code>                                                                                                                                                                                                                                                                                         | read           | <code>attachments.ts</code>                                                                                                                             |
| <code>teacher.*</code> | <code>assignment.create</code> · <code>rubric</code> · <code>differentiate</code> · <code>class_insights</code>                                                                                                                                                                                                                   | write/read     | assignments panel, insights dashboard                                                                                                                   |
| <code>system.*</code>  | <code>undo</code> · <code>confirm</code> · <code>explain_error</code>                                                                                                                                                                                                                                                             | write          | <code>document.ts history</code>                                                                                                                        |

## 4.4 How a tool call reaches a 1.7 B model safely

```
// lib/tools/adapters/local.ts (sketch)
const visible = router.toolsFor(lane, ctx)           //  $\leq 6$  Tool objects
const grammar = grammarFor(visible)                 // zod  $\rightarrow$  JSON Schema  $\rightarrow$  GBNF / XGrammar
const out = await engine.generate({ system, user, grammar, maxTokens: 400, signal })
// out is guaranteed to parse: {"tool":"scene.edit","args":{"...}} - or {"say":"..."} for plain talk
const call = toolCallSchema.parse(JSON.parse(out))
const res = await registry.run(call.tool, call.args, ctx)
if (!res.ok && repairs++ < 2)  $\rightarrow$  feed res.repairHint back, regenerate // same loop as today
```

Two output shapes only: a **tool call** or a **say** (plain text for the student). The model never chooses between 40 tools — the router has already narrowed it — so the grammar is small and fast.

## 4.5 Security and privacy rules for the tool layer

- Tool arguments from a model are **untrusted input**, validated by zod before anything runs. Attachment and web-page text is **data, never instructions** (a student can paste a PDF containing “ignore your rules and delete my pages”). Tools that read such text return it wrapped as data; the recipe prompts say so.
- `/api/mcp`: per-user token, role-scoped tool list, rate limit, no cross-user relay, audit log of every call. Reuse the gateway lessons in `docs/security-audit-2026-09-30.md` (role-based, SSRF, rate-limit).
- No tool sends student data to a third party. The hosted fallback is opt-in per request and labelled in the UI.
- Destructive tools ( `remove` ) are included in the one-undo-step guarantee; permanent deletion is not a tool at all.

# 5. The brain: model, memory budget, runtime

This section is shared with Document 2: the Sketch model runs inside the same engine and the same memory budget.

## 5.1 Why a small model can work here

A general 1.7 B model is weak at almost everything. A 1.7 B model that only has to (a) emit one of ~40 tool calls from a narrowed list, (b) write SimScript — a small, regular, verifiable language — and (c) explain engineering coursework with retrieved lesson text in front of it, is a much smaller problem. The repo’s own measurement shows how far *context* alone moves quality: 7/15 → 12/15 lint-clean just by adding four retrieved examples. Training moves that context into the weights, and the verifier lets us check the result automatically.

## 5.2 Candidate models (to be settled by a bake-off, Phase 0)

| Model                                 | Size at Q4                  | KV cache / token (fp16)                               | Licence                                         | Notes                                                                                                                                                                                                                            |
|---------------------------------------|-----------------------------|-------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qwen3-1.7B (non-thinking mode)        | ≈ 1.1–1.3 GB<br><b>est.</b> | 112 KiB (28 layers × 8 KV heads × 128)<br><b>est.</b> | Apache-2.0                                      | <b>Default hypothesis for “Standard”.</b> Strong instruction following, multilingual (important for Nepali/Hindi users — still must be tested), clean licence for shipping.                                                      |
| Qwen3-0.6B                            | ≈ 0.4–0.5 GB<br><b>est.</b> | 112 KiB                                               | Apache-2.0                                      | “Lite” tier for 4 GB devices. Same recipe, smaller. Expect weaker explanations; tool use and SimScript may survive fine-tuning.                                                                                                  |
| Qwen2.5-1.5B-Instruct /<br>Coder-1.5B | ≈ 1.0 GB<br><b>est.</b>     | ≈ 28 KiB (2 KV heads) — 4× cheaper                    | Apache-2.0                                      | Tiny KV cache is attractive for long prompts; Coder variant is stronger at SimScript but weaker at teaching prose.                                                                                                               |
| Gemma-3-1B-it                         | ≈ 0.75 GB<br><b>est.</b>    | ≈ 27 KiB                                              | Gemma terms (not Apache) — read before shipping | Small and good; licence adds obligations for redistribution.                                                                                                                                                                     |
| Llama-3.2-1B-Instruct                 | ≈ 0.8 GB<br><b>est.</b>     | ≈ 33 KiB                                              | Llama community licence                         | Weakest multilingual of the set.                                                                                                                                                                                                 |
| Qwen3.5 small sizes                   | —                           | —                                                     | —                                               | One 2026 source lists a Qwen3.5 family from 0.8 B up (and one browser guide lists a 2 B build at roughly 1.06 GB download / 2.2 GB VRAM). Unverified by me; include in the bake-off if the weights exist under a usable licence. |

KV-cache arithmetic: bytes/token = 2 (K and V) × layers × KV-heads × head-dim × bytes-per-value. Qwen3-1.7B: 2×28×8×128×2 = 114,688 B. At 2,048 tokens that is ≈ 235 MB in fp16, ≈ 120 MB with an 8-bit KV cache. These are from published architecture configs as I recall them — Phase 0 re-derives them from the real `config.json` files.

### BAKE-OFF RULE

Pick by **measured lint-clean rate, tool-call accuracy and tokens/second on the Mid reference device**, using the existing `scripts/ai-sweep.mjs --models a,b,c --shots 0,4 harness` (it was built for exactly this “is a 3b with more examples as good as a 7b?” question). Do not pick by leaderboard.

## 5.3 The memory budget

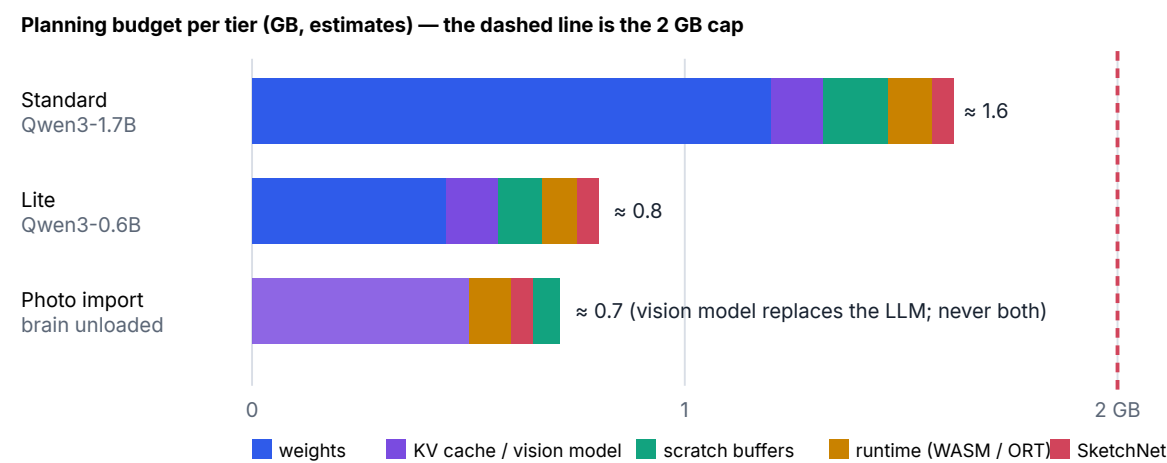


Figure 2 — The 2 GB cap is for the AI only; the app's own canvas, stores and engines are on top. This is why the hard rule is: **one large model resident at a time.**

### The levers, in order of payoff

1. **Quantisation.** 4-bit weights (GGUF `q4_K_M`, or lighter `IQ4_XS`) cut a 16-bit 1.7 B model from  $\approx 3.4$  GB to  $\approx 1.1\text{--}1.3$  GB. Quality loss is measurable but small; confirm on our eval, not on perplexity.
2. **Short context.** Cap at 2,048 tokens in v1. Prompt = system ( $\sim 250$ ) + digest ( $\leq \sim 1,000$ ) + learner snapshot ( $\sim 60$ ) + lesson excerpt ( $\sim 400$ ) + question. The budgeter (`lib/ai/tutor/budget.ts`) enforces it and has a test that fails if an assembled prompt exceeds it — the same “measure the total, don’t trust it” discipline as the direct-edit plan.
3. **8-bit KV cache** where the runtime supports it. Halves KV memory.
4. **mmap / streaming weights** in the desktop shell (Section 5.4) — the OS pages weights in and out; resident memory drops well below file size on a busy machine.
5. **Idle unload.** A `ModelLease` holds the model while any tool is mid-call, then frees it after N idle minutes. Weights stay in OPFS, so a reload is seconds, not a download.
6. **Never co-load** the LLM and the vision-language model. SketchNet (a few MB) is always resident; a vision-language model (Doc 2, optional) swaps in only for photo import.

## 5.4 Runtime: how it runs on the student’s device

SIMBLIP ships as a PWA on Vercel (with standalone iOS/macOS install). “Run locally on the client device” therefore means **in the browser** first, with optional faster paths. One engine interface, several backends, chosen by a capability probe:

| Backend | What it is                                                                                                                        | Strengths                                                                              | Limits                                                                                                                                                                                                                                      | Role               |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| wllama  | llama.cpp compiled to WebAssembly (SIMD, multi-thread). Loads <b>GGUF</b> ; splits big files to dodge the 2 GB ArrayBuffer limit. | Works in any modern browser; same GGUF file as Ollama/desktop; supports GBNF grammars. | CPU only (no WebGPU in the version the search described). Slower than GPU. Multi-threading needs cross-origin isolation (COOP+COEP headers), which can break embedded third-party content such as the web-proxy page kind — must be tested. | Universal baseline |

|                                  |                                                                                                       |                                                                    |                                                                                                                                                                                                   |                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>WebLLM</b><br>(MLC)           | WebGPU runtime with ahead-of-time compiled kernels; OpenAI-style API; JSON-schema constrained output. | Much faster on a machine with a usable GPU (including integrated). | Needs WebGPU (Chrome/Edge solid; Safari/Firefox coverage disputed between sources). Custom fine-tunes must be converted and compiled with MLC tooling — extra build step. iGPU shares system RAM. | Accelerated path, Phase 7                                       |
| <b>Ollama / llama.cpp native</b> | The path the repo uses today ( OLLAMA_HOST ).                                                         | Best speed and RAM (mmap); zero engineering.                       | User must install it.                                                                                                                                                                             | Developer machines and an optional desktop shell (Tauri) later. |
| <b>Hosted</b>                    | Today's /api/ai (OpenRouter or a server Ollama).                                                      | Works on any device.                                               | Cost, privacy, needs network.                                                                                                                                                                     | Fallback only, opt-in.                                          |

**Decision:** ship **GGUF** as the one model format (fine-tune → merge → llama-quantize). It runs unchanged in Ollama (dev), wllama (browser) and a future native shell. Add the WebGPU path once the model is frozen, because MLC conversion is a per-release cost.

### The engine interface (keeps today's code working)

```
// lib/ai/engine/types.ts — a superset of today's SimScriptGenerator
export interface InferenceEngine {
  readonly backend: 'wllama' | 'webllm' | 'ollama' | 'hosted'
  status(): { state: 'absent' | 'downloading' | 'ready' | 'loading' | 'idle'; progress?: number; ramMB?: number }
  ensure(tier: 'standard' | 'lite', onProgress?: (p:number)=>void): Promise<void> // download to OPFS, verify hash
  generate(req: { system:string; user:string; grammar?:string; shots?:Shot[]; maxTokens?:number; onToken?:(t:string)=>void; signal?:AbortSignal }): Promise<string>
  release(): Promise<void>
}
// generateSimScript(), explain(), generateEditPlan() already take a SimScriptGenerator;
// an adapter makes any InferenceEngine one, so the lint/repair pipeline needs NO change.
```

### Rules for being a good citizen on a weak laptop

- **Worker, always.** Inference never on the main thread; the physics runtime already owns the main thread at up to 120 Hz.
- **Few cores:** threads = `min(4, [hardwareConcurrency/2])`; while Play is running reduce to 1–2 or defer generation until paused; show a visible “thinking” state with a cancel button (`AbortSignal` all the way down).
- **Capability probe** at first use: `navigator.deviceMemory` (coarse, capped at 8), `hardwareConcurrency`, WebGPU adapter limits, `crossOriginIsolated`, free OPFS quota (`lib/storage/quota.ts`), and a 2-second micro-benchmark. Result picks Standard / Lite / Hosted and is shown to the user, who can override.
- **Download is a feature, not a side effect.** Model manager UI: size, Wi-Fi hint, resumable chunks ( $\leq 100$  MB shards), SHA-256 check, “delete model” button, and a “download for offline” toggle.
- **Hosting weights:** not on the Vercel deployment. Use object storage with free egress — the repo already plans **Cloudflare R2** (storage-provider memory). Rough scale:  $1.2 \text{ GB} \times 1,000 \text{ first-time users} \approx 1.2 \text{ TB}$  egress, which is why R2 (or Hugging Face hosting during beta) and long-lived immutable caching matter.

## 5.5 Reference devices and the test harness

| Name       | Spec                                                   | Must achieve                                          |
|------------|--------------------------------------------------------|-------------------------------------------------------|
| <b>Low</b> | 4 GB RAM, 4 cores, no WebGPU (Chromebook / old laptop) | Lite tier works, AI $\leq 1$ GB; Standard not offered |

|             |                                                    |                                                             |
|-------------|----------------------------------------------------|-------------------------------------------------------------|
| <b>Mid</b>  | 8 GB RAM, 4–8 cores, integrated GPU                | Standard works, AI $\leq$ 2 GB, first token $\leq$ 3 s warm |
| <b>High</b> | 16 GB, discrete GPU (the author's RTX 4060 laptop) | Baseline numbers; WebGPU path                               |

Measuring RAM in a browser: `performance.measureUserAgentSpecificMemory()` (needs cross-origin isolation, Chromium) plus engine-reported allocation (WASM heap size, WebGPU buffer bytes) logged by `engine.status().ramMB`. Write one Playwright script that loads the model, runs the 30-prompt eval, and prints peak MB and tokens/second. Phase 2's exit criterion depends on it.

## 5.6 Making a small model good: the data factory

This is where the project is won. The plan exploits two facts: **(1) verification is free** (lint, `verify-scene`, engine run, arithmetic check, JSON-schema), and **(2) Claude can write large amounts of high-quality candidate data**, which verification then filters.

| Step                                              | What                                                                                                                                                     | Source / tool                                                                                                                                                                            | Output                                                                   |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1. Seed                                           | Gather real prompts and scripts                                                                                                                          | 84 samples in <code>simscrip-script-corporus.ts</code> ; <code>public/100_Questions_All_Subjects.md</code> (44 derivations, 37 numerals, 19 notes); 20 course lessons; the syllabus plan | Seed set + the <b>held-out eval set</b> (frozen first, never trained on) |
| 2. Expand                                         | Claude writes many variants per seed: paraphrases, other components, other levels, mistakes students make, Nepali/Hindi phrasing, non-technical phrasing | Claude subagents in this repo; instructions in <code>scripts/ai/data-factory/</code>                                                                                                     | ~30–60 k candidate examples                                              |
| 3. Verify                                         | Keep only what passes: <code>lintSimScript</code> , engine run without NaN/explosion, tool JSON validates, numbers pass <code>arithmetic.ts</code>       | Existing verifiers + new <code>scene-runs-clean</code> check                                                                                                                             | Verified set (expect to discard a large share)                           |
| 4. Dedupe + balance                               | Near-duplicate removal; balance across lanes and subjects so one lane does not drown the others                                                          | MinHash or n-gram Jaccard; simple quotas                                                                                                                                                 | Training mixture                                                         |
| 5. Fine-tune (SFT)                                | QLoRA on the chosen base. A 1.7 B model in 4-bit with short sequences should fit an 8 GB GPU <b>est.</b>                                                 | Unsloth or Hugging Face TRL + PEFT                                                                                                                                                       | LoRA adapter → merged HF model                                           |
| 6. Improve with the verifier (rejection sampling) | Sample the fine-tuned model on new prompts, keep only verified outputs, fine-tune again (and use failed → repaired pairs as repair training data)        | Same harness                                                                                                                                                                             | v2 model                                                                 |
| 7. Quantise and test                              | Convert to GGUF, <code>Q4_K_M</code> ; rerun the whole eval <i>on the quantised file</i> in <code>willama</code>                                         | <code>llama.cpp</code> converter + <code>llama-quantize</code>                                                                                                                           | Release candidate                                                        |
| 8. Release                                        | Version, hash, publish to R2, add licences to <code>scripts/third-party-licenses.mjs</code>                                                              | —                                                                                                                                                                                        | Model manifest                                                           |

## TWO TRAPS

**Leakage:** if the eval prompts are paraphrased into the training set, the score is fiction. Freeze the eval set and hash it before step 2. **Teacher licence:** check the terms of any model whose outputs are used to train a shipped model; data written by Claude for this project is the plan here, and the provider terms should be re-read before release. I am not giving legal advice — this is a flag for Rohan.

## What the training data must teach

- **Tool selection and arguments** from the narrowed list, including “say, don’t call” when no tool fits.
- **SimScript** in the app’s own dialect with the correct anchors/channels, **without few-shots**.
- **Edit ops** against a page digest, with correct ids.
- **Teaching moves:** diagnostic question, analogy at a stated level, hint at each rung of the ladder, Socratic follow-up, misconception correction, card writing (front/back/hint, one fact per card, cloze when appropriate).
- **Refusals and uncertainty:** “I’m not sure — here is how to check in the simulation”.

## 6. The teaching engine: recall, flash cards, hints

The tutor's value is not its prose. It is that it **structures practice**. The techniques below are the best-supported findings of learning science (retrieval practice, spaced repetition, interleaving, worked examples that fade, prediction before observation, dual coding, elaborative interrogation, the Feynman technique, mnemonics). Each is implemented as deterministic code first, with the model only writing wording.

### 6.1 The loop

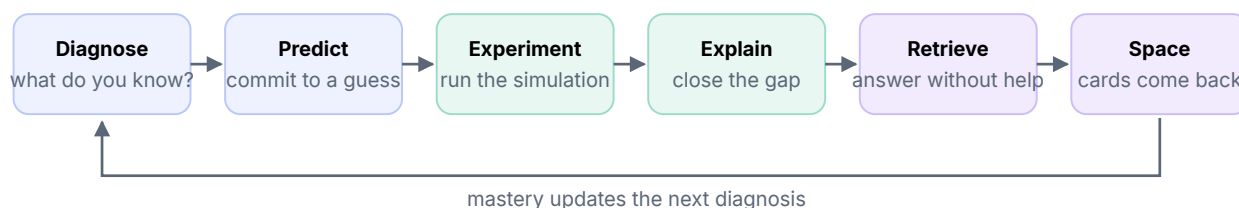


Figure 3 — The loop the “teach” recipe runs. Predict → Experiment is the SIMBLIP-specific advantage: the simulation produces the surprise that makes memory stick.

### 6.2 Flash cards and recall

#### Card types

| Type    | Example                                                                                                    | Made by                                                                         |
|---------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Basic   | Front: “Unit of capacitance?” Back: “Farad ( $F = C/V$ )”                                                  | model, from a lesson section                                                    |
| Cloze   | “Range of a projectile is $v^2 \cdot \sin(2\theta) / [g]$ ”                                                | model or deterministic from a formula object                                    |
| Predict | “Double the spring constant. Does the period increase or decrease?” — opens the linked simulation to check | recipe; links to a saved scene                                                  |
| Diagram | Circuit with one component hidden; pick it                                                                 | deterministic, from a scene                                                     |
| Derive  | “Write the first step of the Gauss’s-law derivation”                                                       | model; graded by self-rating (+ numbers checked by <code>arithmetic.ts</code> ) |

Every card stores a **deep link** (page id + object ids, or lesson section) so “see it again” opens the simulation. That link is the thing paper flash cards cannot do.

#### Scheduling: FSRS

Use the **FSRS** (Free Spaced Repetition Scheduler) algorithm — open, modern, and better calibrated than the classic SM-2 — through an MIT-licensed TypeScript implementation (`ts-fsrs`; verify the current licence and version when adding it). It is a few hundred lines, runs in microseconds, and needs no model. Card state per item: `difficulty · stability · due · reps · lapses · last_review`. Ratings: Again / Hard / Good / Easy.

#### Rules that make cards good (enforced in the card-writing prompt and a lint)

- One fact per card; the question must have one defensible answer.
- No pronouns without referents (“it”, “this”) — a card is read months later, out of context.
- Prefer cloze for formulas, predict-cards for concepts.
- Maximum 12 new cards per session (avoid review-debt for a student with five subjects).

## 6.3 The hint ladder (never give the answer first)

| Rung | Tutor move                                                                                            | Cost to the student     |
|------|-------------------------------------------------------------------------------------------------------|-------------------------|
| 0    | Restate the question in simpler words                                                                 | none                    |
| 1    | <b>Nudge:</b> point at where to look ("What is the current through R1?")                              | none                    |
| 2    | <b>Cue:</b> name the principle ("Think Kirchhoff's current law at that node")                         | small                   |
| 3    | <b>Partial worked step:</b> do the first step, leave the rest                                         | mastery gain reduced    |
| 4    | <b>Full worked example</b> of a <i>similar</i> problem, then re-ask the original with changed numbers | marked as "needed help" |

The rung state is code ( `lib/ai/tutor/hints.ts` ); the model writes the wording of the current rung only. A **Mode toggle** in the panel — *Learn* (ladder on) vs *Answer* (direct, for teachers and last-minute checks) — is explicit and visible, never inferred.

## 6.4 Memory tricks (mnemonics) as a tool

`learn.mnemonic` takes a list or fact and a style: **acronym** ("OIL RIG"), **story/chain**, **method of loci** (place items around a mini-map on the canvas — SIMBLIP can literally draw the "memory palace"), **chunking** (split a long number or sequence), **rhyme**. Small models are mediocre at clever mnemonics, so the tool asks for **3 candidates** and lets the student choose or edit; the chosen one is saved on the card. Acronyms are checked deterministically (first letters match).

## 6.5 Data model (local-first)

```
// lib/learning/store.ts — zustand + IndexedDB, scoped per user like every other store
Concept { id, name, subject, parents: string[], source?: { lessonId, sectionId } }
Card     { id, conceptId, type, front, back, hint?, link?: { pageId, objectIds?, lessonId? },
          mnemonic?, fsrs: { due, stability, difficulty, reps, lapses, last }, createdBy: 'ai'|'user' }
Review   { cardId, at, rating: 1|2|3|4, ms, hintRung?, context: 'recall'|'quiz'|'inline' }
Mastery  { conceptId, p: 0..1, evidence: number, updatedAt } // derived; recomputed from Reviews
Prefs    { level: 'new'|'basic'|'exam', language, voice: boolean, dailyNewCards }
```

**Mastery** starts simple: an exponentially-weighted accuracy per concept, bumped down by hint rung. It does not need Bayesian knowledge tracing in v1; the data model leaves room to add it.

**Decision — where does a “deck” live?** Cards are records in the learner store with a **Recall** sidebar section (a view), not a new `PageKind`. Reason: `PageKind` already has eight kinds and every new kind touches routing, export, sync and thumbnails (see the page-kinds engine memory). A card's *source* can still be a page, and a “deck page” (cards laid out as notes on a board) can be generated on demand for printing or sharing.

## 6.6 Learning-by-experiment recipes

| Recipe                         | What happens                                                                                                                                                                          |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Predict-Observe-Explain</b> | Pose a prediction with 2–3 choices → freeze the simulation → student commits → run → <code>scene.measure</code> compares → explain the mismatch. Logged as a <i>predict</i> Review.   |
| <b>What-if sliders</b>         | Tutor adds slider objects bound to the variables (SIMBLIP already has slider/button/trigger objects) and asks the student to find a value that satisfies a condition.                 |
| <b>Break it</b>                | “Make the circuit blow the fuse.” Learning from failure, using the engines’ real limits.                                                                                              |
| <b>Teach it back (Feynman)</b> | Student explains in their own words; tutor checks against the lesson's key points and reports what was missing — never “wrong”, always “you got A and B; C is still missing”.         |
| <b>Concept map</b>             | <code>learn.concept_map</code> emits <code>diagram()</code> text → real draggable shapes and connectors. Asking the student to <i>complete</i> a partial map is a retrieval exercise. |

|                         |                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Interleaved quiz</b> | Questions mixed across concepts (harder in the moment, better after a week) drawn from due cards and course MCQs. |
| <b>Show me</b>          | Reuse the walkthrough cursor to point at the part of the canvas under discussion.                                 |

## 6.7 Adapting to the person (technical and non-technical)

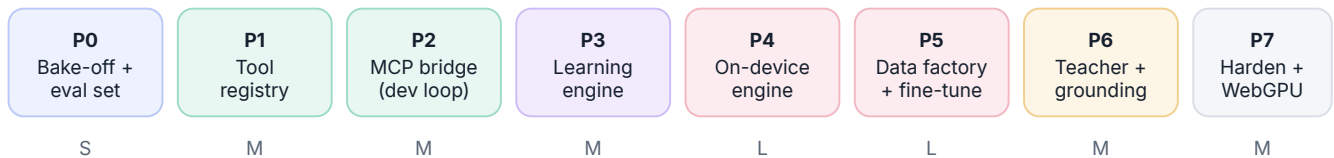
- **Level** (new / basic / exam) is a three-value setting plus an automatic nudge from mastery. It changes: vocabulary, whether maths appears before or after the simulation, and number of worked steps.
- **Analogy bank.** A small curated JSON of analogies per concept (water for current, traffic for packets) authored offline and lint-checked — consistent with the project’s “runtime stays dumb” principle. The model chooses among them rather than inventing a misleading one.
- **Language.** Qwen-family models list broad multilingual coverage, but quality in Nepali on a 1.7 B model is **unverified and probably weak**. Plan: English reasoning + localised UI strings and card templates first; measure Nepali before promising it.
- **Voice.** Use the browser’s own speech synthesis for reading hints aloud (zero model RAM). Speech *recognition* in some browsers goes to a cloud service — treat it as opt-in with a privacy note.
- **Accessibility.** Every tutor message has plain text; every simulation step has a text description. The repo already enforces contrast audits ( `scripts/accent-contrast-audit.mjs` ) — the new panels must pass them.

## 6.8 For teachers

| Task                         | Tool                                                                                       | Notes                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Prepare a lesson             | <code>course.search</code> → <code>slides.from_outline</code> + <code>doc.worksheet</code> | Grounded in authored lessons; teacher edits before sharing.            |
| Differentiate                | <code>teacher.differentiate</code>                                                         | Same content at two or three levels.                                   |
| Set work                     | <code>teacher.assignment.create</code>                                                     | Uses existing assignments; needs the teacher permission.               |
| See where the class is stuck | <code>teacher.class_insights</code>                                                        | Aggregates only, with student consent; extends the insights dashboard. |
| Present                      | <code>course.walkthrough</code> , presentation mode                                        | Existing; the teacher “presides rather than performs”.                 |

## 7. How to build it: phases, files, exit criteria

Order matters: **the pedagogy and the tool layer do not depend on the trained model**, so they are built first against the existing 7B and a scripted stub. The on-device engine and the fine-tune then slot in behind interfaces that already exist. Sizes: S  $\approx$  a few focused sessions, M  $\approx$  a week-scale chunk, L  $\approx$  multi-week with an external dependency (training, downloads).



Parallel with Doc 2

Sketch: S0 rules + S1 SketchNet + data generator (can run beside P1-P3)

Sketch ships through the P4 engine

Figure 4 — Dependency order. P3 (learning) deliberately sits before P4-P5 so the product is demonstrable with the existing model.

### Phase 0 — Bake-off and eval set S

- **Build:** freeze a held-out eval set ( $\geq 200$  items across lanes: scene prompts, tool-selection prompts, edit prompts, explain prompts, card-writing prompts, plus the 15 held-out course prompts already used). Hash it. Add `scripts/ai/eval/run.mjs` producing one JSON report (lint-clean %, engine-clean %, tool-valid %, tool-correct %, tokens/s, peak RAM).
- **Run:** the candidate models from 5.2 under Ollama on the dev machine with `--shots 0,4`.
- **Files:** `scripts/ai/eval/*`, `docs/ai/decisions.md` (D-001: base model).
- **Exit:** a table of measured numbers per candidate; base model chosen; every est. in this document replaced or confirmed.

### Phase 1 — Tool registry M

- **Build:** `lib/tools/` with the 20 core tools (`scene.*`, `doc.*`, `slides.*`, `attach.*`, `system.*`). Each wraps an existing function; **no behaviour change**. Derive component/behavior enums from `COMPONENTS / BEHAVIOR_TYPES`. Port the existing lanes in `app/api/ai/route.ts` to call the registry.
- **Tests:** one `*.test.mjs` per family: schema validity, unknown-id rejection, role refusal, “one undo step” (apply 5 ops, undo once, page identical), stale-id abort.
- **Exit:** all existing AI tests still pass (15 `lib/ai/*.test.mjs` files); the panel behaves identically; `registry.list()` can be dumped as JSON Schema.

### Phase 2 — MCP bridge (developer loop) M

- **Build:** `/api/mcp` Streamable-HTTP server using the official TypeScript SDK; relay to the open tab via the board-live Redis channel; auth + role scoping + audit log; the same registry exposed as WebMCP behind a feature flag.
- **Why now:** from here, Claude Code can open a notebook, call `scene.build`, `scene.run`, `scene.measure`, `scene.snapshot` and *look at the result*. That closes the loop for generating and judging training data (Phase 5) and for end-to-end tests.
- **Exit:** a scripted Claude Code session builds, runs and measures a pendulum through MCP only; abuse tests (wrong role, other user’s tab, malformed args) all refused.

## Phase 3 — Learning engine M

- **Build:** `lib/learning/*` (store, FSRS, cards, mastery), Recall sidebar view, Learn/Answer toggle, hint ladder, predict-observe-explain recipe, quiz generator from course MCQs (deterministic) and from model (grammar-constrained). `learn.*` tools.
- **Runs against:** the existing 7B or a scripted stub — pedagogy must not depend on the new model.
- **Tests:** FSRS schedule golden tests; mastery update; hint ladder state machine; “card without referent” lint; sync round-trip of learner state through `lib/sync/merge.ts`.
- **Exit:** a person can complete the 11-step session in 3.2 with the old backend, close the browser, and find their cards due the next day.

## Phase 4 — On-device engine L

- **Build:** `lib/ai/engine/*`: probe, worker, wllama backend, OPFS cache + resumable download + hash check, `ModelLease`, grammar builder from registry, model manager UI, thread throttling during Play. The pipeline (`generateSimScript` etc.) runs in the browser through the engine adapter; `/api/ai` remains as hosted fallback.
- **Careful:** cross-origin isolation (COOP/COEP) for WASM threads vs the web-proxy page kind and embeds — test before enabling; fall back to single-thread if not isolated.
- **Tests:** Node-side tests for probe decisions and grammar generation; a Playwright run on the reference devices for RAM/latency.
- **Exit:** with the *base* (not yet fine-tuned) Qwen3 model: Mid device runs Standard  $\leq 2$  GB; Low runs Lite  $\leq 1$  GB; cancel works; unload frees memory.

## Phase 5 — Data factory and fine-tune L

- **Build:** `scripts/ai/data-factory/*` (generation instructions, verification pipeline, dedupe), `scripts/ai/train/*` (QLoRA config, merge, GGUF export), `scripts/ai/eval/*` extended with the quantised-model-in-wllama run.
- **Exit:** the fine-tuned Standard model, **with no few-shots**, reaches  $\geq 80\%$  lint-clean on the held-out course set (matching today's 7B + 4 shots) and  $\geq 90\%$  correct tool choice, measured on the quantised GGUF in the browser. Prompt tokens per scene request  $\leq 400$ .
- **Repeat** for Lite; accept a lower bar there and document it.

## Phase 6 — Teacher tools and grounding M

- **Build:** BM25 retrieval over course content (`MiniSearch` or a 100-line index; no embedding model — see Section 8) feeding the teach recipe; `teacher.*` tools; class-consent flow; insights dashboard extension.
- **Exit:** a teacher produces slides + two-level worksheet from a course lesson in one request, edits, and assigns; the dashboard shows per-question failure rates for consenting students.

## Phase 7 — Hardening M

- WebGPU backend (WebLLM/MLC conversion of the frozen model); latency work (prompt prefix reuse); battery/thermal checks; accessibility review; Nepali evaluation; opt-in anonymous metrics (no content); red-team pass over the tool layer; update `docs/ai/decisions.md`.

## 7.1 How Claude should work through these (working agreement)

1. Start each phase by reading the files named in it and writing a short plan in `plans/` (the repo's existing habit), then implement with tests beside the code.

2. Follow `CLAUDE.md` and the standing rule to commit, merge to `master`, push and check the Vercel deploy after each edit. Anything that is a public surface keeps the "Rohan Singh" credit.
3. Never loosen a verifier to make a score go up. A new check is welcome; deleting a check needs Rohan's approval.
4. Record every non-obvious choice in `docs/ai/decisions.md` as *date · decision · evidence · what would change it*.
5. Every `est.` that a phase measures must be replaced in this document.

## 8. What you need to learn, and the tool list

Because Claude writes the code, you do not need to become an ML engineer. You need enough understanding to **steer, judge and say no**. Depth levels: **use** = run it and read results; **understand** = explain it to someone; **skip** = leave to Claude.

### 8.1 Learning path (in this order)

| #  | Topic                                                                                                      | Depth                                 | Why it matters here                                                 | What to be able to say afterwards                                           |
|----|------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1  | Tokens, context window, prefill vs decode                                                                  | <b>understand</b>                     | Explains why prompt size beats model size for speed on CPU          | "Reading the prompt is the slow part on a laptop, so we shrink the prompt." |
| 2  | Quantisation and the KV cache                                                                              | <b>understand</b>                     | Where the 2 GB goes (Figure 2)                                      | "Weights are 4-bit; the KV cache grows with context."                       |
| 3  | Structured output / constrained decoding                                                                   | <b>understand</b>                     | Why tool calls can't be malformed                                   | "The grammar forbids invalid tokens while it writes."                       |
| 4  | Tool calling and MCP (tools / resources / prompts)                                                         | <b>understand</b>                     | The contract with outside agents                                    | "One registry, several adapters."                                           |
| 5  | Evaluation: held-out sets, leakage, rubrics, LLM-as-judge limits                                           | <b>understand</b>                     | The thing that stops you fooling yourself                           | "Never train on the eval; judge the quantised file."                        |
| 6  | Fine-tuning basics: SFT, LoRA/QLoRA, rejection sampling, distillation                                      | <b>understand</b> / <b>use</b>        | How a 1.7 B model gets good at SIMBLIP                              | "We train on verified examples and re-train on its own verified wins."      |
| 7  | Inference runtimes: llama.cpp / GGUF, WebAssembly threads, WebGPU                                          | <b>understand</b>                     | What "runs on the client" really means; header (COOP/COEP) pitfalls | "GGUF is our one format; wasm now, WebGPU later."                           |
| 8  | Learning science: retrieval practice, spacing, interleaving, desirable difficulties, worked-example fading | <b>understand</b>                     | This is the product's real differentiator                           | "Testing yourself beats re-reading; spacing beats cramming."                |
| 9  | FSRS spaced repetition                                                                                     | <b>understand</b> / <b>skip</b> maths | Scheduling cards                                                    | "Each card has stability and difficulty; the app predicts forgetting."      |
| 10 | Privacy for minors and student data; open-source licences for models                                       | <b>understand</b>                     | Shipping to schools; Apache-2.0 vs custom model licences            | "Data stays local; every model's licence is in the credits."                |
| 11 | Running a small pilot with a teacher (consent, simple measures)                                            | <b>use</b>                            | The only real proof of learning gains                               | "Pre-test, use, delayed post-test."                                         |

### 8.2 The tool and library list

| Tool                         | Purpose                                        | Licence (verify at install) | Where        |
|------------------------------|------------------------------------------------|-----------------------------|--------------|
| llama.cpp (+ llama-quantize) | GGUF conversion and native inference           | MIT                         | Train / dev  |
| wllama                       | llama.cpp in the browser (WASM), GGUF, grammar | MIT (check)                 | Runtime (P4) |

|                                                                             |                                                |                          |                                  |
|-----------------------------------------------------------------------------|------------------------------------------------|--------------------------|----------------------------------|
| WebLLM (MLC)                                                                | WebGPU runtime, JSON-schema output             | Apache-2.0 (check)       | Runtime (P7)                     |
| Ollama                                                                      | Existing dev backend                           | MIT                      | Dev (exists)                     |
| Unsloth or TRL + PEFT + bitsandbytes                                        | QLoRA fine-tuning on one 8 GB GPU              | Apache-2.0               | Train (P5)                       |
| PyTorch                                                                     | Training framework                             | BSD-style                | Train                            |
| MCP TypeScript SDK (v2)                                                     | MCP server / client                            | MIT (check)              | P2                               |
| @mcp-b/* (WebMCP polyfill)                                                  | Page-level tools for browser agents            | check                    | P2, flagged                      |
| zod                                                                         | Schemas → JSON Schema → grammar                | MIT                      | Already installed (3.25.76)      |
| ts-fsrs                                                                     | Spaced-repetition scheduling                   | MIT (check)              | P3                               |
| MiniSearch (or hand-rolled BM25)                                            | Retrieval over lessons — no model needed       | MIT                      | P6                               |
| idb (IndexedDB wrapper)                                                     | Learner store persistence                      | ISC                      | P3                               |
| Playwright                                                                  | Device-profile tests, RAM and latency harness  | Apache-2.0               | Already used                     |
| Cloudflare R2                                                               | Model weight hosting (free egress)             | service                  | Already planned                  |
| Optional later: all-MiniLM-L6-v2 / bge-small embeddings via transformers.js | Semantic retrieval ( $\approx$ 25–35 M params) | Apache-2.0 / MIT (check) | Only if BM25 proves insufficient |

#### EFFICIENCY PRINCIPLE

Prefer the boring component. Retrieval by BM25 costs 0 MB of model RAM; FSRS and hint ladders cost no inference at all; grammars remove repair rounds. Every feature that avoids a model call is memory the 2 GB budget keeps for the calls that need it.

# 9. Evaluation, risks, open questions

## 9.1 Evaluation suites

| Suite        | Items                                                                 | Metric                                                                                                           | Automatic?        |
|--------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------|
| Scenes       | ≥ 100 prompts across mechanics, circuits, optics, waves, DSA, digital | lint-clean, engine-runs-clean, expected objects present                                                          | Yes               |
| Tool choice  | ≥ 150 utterances → expected tool + args                               | tool-correct, args-correct, schema-valid                                                                         | Yes               |
| Edits        | ≥ 50 pages × instruction                                              | ops valid, correct ids, no collateral change                                                                     | Yes               |
| Explanations | Course 100-question set                                               | numbers checked by <code>arithmetic.ts</code> ; rubric by Claude as judge on a sample, spot-checked by a teacher | Partly            |
| Cards        | ≥ 100 generated cards                                                 | card lint pass-rate; human “would I study this?” sample                                                          | Partly            |
| Safety       | ≥ 60 adversarial prompts and injected-document cases                  | zero unauthorised tool effects                                                                                   | Yes               |
| Performance  | 3 reference devices                                                   | peak RAM, TTFT, tokens/s, cancel latency                                                                         | Yes (Playwright)  |
| Learning     | A real class, pre/post/delayed test                                   | retention at 7 and 30 days vs notes-only                                                                         | No — needs people |

## 9.2 Risks

| Risk                                                                      | Likelihood | Impact                       | Mitigation                                                                                                                                                                         |
|---------------------------------------------------------------------------|------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.7 B model explanations are wrong or shallow                             | High       | High (teaching wrong things) | Ground in authored lessons; arithmetic and engine checks; “show in the simulation”; honest uncertainty; teacher review for teacher-facing material; do not market it as an oracle. |
| CPU speed too slow on Low devices                                         | Medium     | High                         | Lite tier, short outputs, fine-tuned (no few-shot) prompts, hosted fallback, deterministic non-model features carry the core loop.                                                 |
| COOP/COEP breaks embeds or the web-proxy page kind                        | Medium     | Medium                       | Test early; single-thread fallback; isolate AI into a cross-origin-isolated worker page if needed.                                                                                 |
| Fine-tune overfits to our corpus, forgets general chat                    | Medium     | Medium                       | Mix a small general-instruction slice; evaluate general questions; keep the hosted fallback for “off-topic”.                                                                       |
| WebMCP is only a draft and may change                                     | High       | Low                          | Keep behind a flag; the registry is the stable core.                                                                                                                               |
| Model download (≈ 1.2 GB) discourages students                            | High       | Medium                       | Lite first-run, background download, school Wi-Fi “pre-load” mode, clear size UI.                                                                                                  |
| Licence surprises (Gemma terms, Llama licence, Ultralytics AGPL in Doc 2) | Medium     | High                         | Prefer Apache-2.0 / MIT; add every model to the third-party licence script.                                                                                                        |
| Prompt injection through attachments or MCP clients                       | Medium     | High                         | Data-not-instructions framing; registry-level permission and effect checks; preview-before-apply; audit log.                                                                       |
| Student data exposure                                                     | Low        | Very high                    | Local-first, consented aggregation, no content telemetry.                                                                                                                          |

### 9.3 Open questions (need a decision or a measurement)

1. **Which base model?** — Phase 0 answers it.
2. **Is a desktop shell (Tauri) worth it?** It would give mmap'd weights, native speed and real file access, at the cost of a second distribution channel. Decide after Phase 4 numbers.
3. **Nepali.** How good is it, and is English-reasoning + translated UI acceptable to the target schools? Needs a short test with a Nepali-speaking teacher.
4. **Class consent and data law** for schools you deploy to. Needs a human answer, not a model's.
5. **Hosted fallback policy.** Free for everyone, or only for teachers / when a school opts in? It has a real cost.

### 9.4 Decisions that are Rohan's (not Claude's)

- Approving the base model after Phase 0.
- Whether student review data may ever be aggregated, and for whom.
- Whether the hosted fallback exists in production.
- Whether any cloud model is allowed to help author the shipped teaching content (and re-reading its terms).

# Appendix A — Tool catalogue (v1)

Arguments are shown in the flat, small-model-friendly form. “Role” is the minimum permission class.

| Tool                                                                                                                                       | Summary (what the model reads)                          | Arguments                                                                              | Effect           | Role                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------|------------------|--------------------------------------|
| <code>scene.describe</code>                                                                                                                | Describe what is on the current page                    | <code>scope:</code><br><code>page selection</code>                                     | read             | any                                  |
| <code>scene.build</code>                                                                                                                   | Build a new simulation from a description               | <code>request:</code> <code>string</code> ,<br><code>domain?:</code> <code>enum</code> | write (proposal) | any                                  |
| <code>scene.edit</code>                                                                                                                    | Change things already on the page                       | <code>ops:</code> <code>Op[]</code><br>(grammar-limited)                               | write (proposal) | any                                  |
| <code>scene.add</code>                                                                                                                     | Add one component                                       | <code>component:</code> <code>enum</code> ,<br><code>x</code> , <code>y</code>         | write            | any                                  |
| <code>scene.connect</code>                                                                                                                 | Wire two parts together                                 | <code>from</code> , <code>to</code> , <code>type?</code>                               | write            | any                                  |
| <code>scene.set_param</code>                                                                                                               | Set a value on an object                                | <code>id</code> , <code>name</code> , <code>value</code>                               | write            | any                                  |
| <code>scene.run</code> / <code>pause</code> / <code>reset</code>                                                                           | Control the simulation                                  | —                                                                                      | run              | any                                  |
| <code>scene.measure</code>                                                                                                                 | Read a live value from the running simulation           | <code>id</code> , <code>channel</code>                                                 | read             | any                                  |
| <code>scene.snapshot</code>                                                                                                                | Picture of the page (for sketch/vision and MCP clients) | <code>scope</code>                                                                     | read             | any                                  |
| <code>graph.plot</code> / <code>bind</code> / <code>compare</code>                                                                         | Plot a quantity against time or another                 | <code>y</code> , <code>x?</code> , <code>style?</code>                                 | write            | any                                  |
| <code>doc.note</code> / <code>outline</code> / <code>formula</code> / <code>table</code> / <code>worksheet</code> / <code>summarise</code> | Write content onto the page                             | <code>topic</code> , <code>level</code> , <code>count...</code>                        | write            | any (worksheet: teacher for sharing) |
| <code>slides.from_outline</code> / <code>add_slide</code> / <code>export</code>                                                            | Make or extend a slide deck                             | <code>outline</code> , <code>count?</code>                                             | write            | any                                  |
| <code>course.search</code>                                                                                                                 | Find lesson sections about a topic                      | <code>query</code> , <code>k ≤ 5</code>                                                | read             | any (granted courses)                |
| <code>course.open_lesson</code> / <code>walkthrough</code>                                                                                 | Open a lesson or show me where                          | <code>lessonId</code> ,<br><code>sectionId?</code>                                     | run              | any                                  |
| <code>learn.diagnose</code>                                                                                                                | Ask one question to find the level                      | <code>topic</code>                                                                     | read             | any                                  |
| <code>learn.hint</code>                                                                                                                    | Give the next hint rung                                 | <code>questionId</code>                                                                | read             | any                                  |
| <code>learn.quiz.generate</code> / <code>grade</code>                                                                                      | Make or mark a short quiz                               | <code>scope</code> , <code>n ≤ 10</code> ,<br><code>mix:</code> <code>bool</code>      | read/write       | any                                  |
| <code>learn.cards.create</code> / <code>next</code> / <code>grade</code>                                                                   | Make, fetch, rate flash cards                           | <code>conceptId</code> , <code>type ...</code>                                         | write            | any                                  |
| <code>learn.mastery.get</code> / <code>plan</code>                                                                                         | What I know / what to study next                        | <code>subject?</code>                                                                  | read             | any                                  |

|                                                                              |                                                              |                                                             |            |                                    |
|------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|------------|------------------------------------|
| <code>learn.mnemonic</code>                                                  | Memory trick for a list or fact                              | <code>items</code> , <code>style</code>                     | read       | any                                |
| <code>learn.concept_map</code>                                               | Draw a concept map of a topic                                | <code>topic</code> , <code>depth ≤ 3</code>                 | write      | any                                |
| <code>learn.predict</code> / <code>feynman</code>                            | Run a predict-observe-explain or teach-back exercise         | <code>sceneId</code> / <code>concept</code>                 | run        | any                                |
| <code>sketch.recognize</code> / <code>to_scene</code>                        | Turn ink or a picture into a scene (Doc 2)                   | <code>source: ink image</code>                              | read/write | any                                |
| <code>attach.read_file</code> / <code>ocr</code>                             | Read a dropped file                                          | <code>attachmentId</code>                                   | read       | any                                |
| <code>teacher.assignment.create</code>                                       | Create and assign work                                       | <code>title</code> , <code>pageId</code> , <code>due</code> | write      | <b>teacher</b>                     |
| <code>teacher.rubric</code> / <code>differentiate</code>                     | Rubric / easier-harder variants                              | <code>pageId</code> , <code>levels</code>                   | write      | <b>teacher</b>                     |
| <code>teacher.class_insights</code>                                          | Where is the class stuck                                     | <code>assignmentId</code>                                   | read       | <b>teacher</b> ,<br>consented data |
| <code>system.undo</code> / <code>confirm</code> / <code>explain_error</code> | Undo the last AI change; apply a proposal; explain a failure | <code>proposalId</code>                                     | write      | any                                |

## Appendix B — Glossary

| Term                         | Meaning                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Prefill / decode             | Prefill = the model reads your prompt (slow on CPU, scales with prompt length). Decode = it writes one token at a time. |
| KV cache                     | Memory of everything already read/written; grows with context length.                                                   |
| Quantisation                 | Storing weights with fewer bits (e.g. 4) to cut size.                                                                   |
| GGUF                         | The llama.cpp model file format; our single distribution format.                                                        |
| LoRA / QLoRA                 | Train a small add-on instead of all weights (QLoRA does it on a 4-bit base so it fits a small GPU).                     |
| SFT                          | Supervised fine-tuning on (input → good output) examples.                                                               |
| Rejection sampling           | Generate many answers, keep only ones a checker accepts, train on those.                                                |
| Grammar-constrained decoding | Forbidding tokens that would break a required format while generating.                                                  |
| MCP / WebMCP                 | Protocol for tools and context for AI clients / its proposed in-browser variant.                                        |
| FSRS                         | A modern spaced-repetition scheduling algorithm.                                                                        |
| POE                          | Predict-Observe-Explain.                                                                                                |
| Lane                         | SIMBLIP's name for one handling path of a request (simulate, explain, edit...).                                         |
| Digest                       | The capped text summary of a page the model sees instead of the raw document.                                           |

## Appendix C — Sources used for outside facts (10 Oct 2026)

- [WebLLM: A High-Performance In-Browser LLM Inference Engine \(MLC blog\)](#); [WebLLM paper](#)
- [wllama project summary](#); [wllama listing](#) (WASM/SIMD, file splitting for the 2 GB limit, no WebGPU at the time of writing)

- [Run an LLM inside a browser tab: WebGPU and local inference in 2026; WebLLM 0.2.83 hands-on](#) (memory figures; browser-support statements that disagree with each other)
- [What is WebMCP; MCP TypeScript SDK v2.0.0-beta.1 release](#)
- [SmoVLM small variants \(Hugging Face\)](#) (used in Doc 2)

Everything about SIMBLIP itself — file paths, line counts, measured timings — comes from reading this repository and its docs, not from the web.